Problem F
Question 4: Historical Decomposition and Future Trends
Primary benchmark-space scenario forecast with weak bridge retained only as a directional cross-check

Final synchronized technical report

24 September 2026

Numerical source of truth. Every number, table and plotted quantity in this report is synchronized to the latest successful `python run_all.py` execution supplied by the team. Older report values are superseded.

Abstract

Q4 remains the weakest inferential stage and is presented with explicit specification and bridge sensitivity. The six-benchmark Average is exactly reproducible, and repository-deduplicated C8 validation remains strong. The primary historical model on 63 matched models in 22 families gives $\beta_C = 4.6796$, $\beta_t = 4.1279$, $R^2 = 0.4985$; family fixed effects raise β_C to 9.3332 and flip β_t to -1.9478. With dynamically loaded $g_C = 0.534026$, the scale-associated point share is 37.7% with a 4.6–71.7% bootstrap interval. The headline forecast is directly in benchmark-score space from a documented baseline 51.2313 and gives 56.61 at 12 months and 61.99 at 24 months under a 50% compute-growth slowdown. C6 Loss-to-Benchmark bridges are weak across families and are retained only as directional cross-checks.

Contents

1 Capability metric and C8 validation 1

2 Historical C3 context 1

3 Historical decomposition and specification sensitivity 2

4 Compute-frontier trend and documented baseline 3

5 Primary 12/24-month benchmark-space forecast 4

6 Loss-to-Benchmark bridges: direction-only diagnostics 5

6.1 Ridge variant sensitivity 6

7 Known limitations - honest disclosure 7

8 Final Q4 answer 7

1 Capability metric and C8 validation

Capability is the equal-weight mean of IFEval, BBH, MATH, GPQA, MUSR and MMLU-PRO. Across 4,576 C2 rows, recomputation differs from the supplied Average by at most 1.421×10^{-14}.

C8 contains 1,860 aggregate directories; 1,858 repository keys intersect C2. The join has 1,898 rows before de-duplication and 1,858 after repository de-duplication. On the deduplicated matches,

$$r = 0.9863, \qquad \rho = 0.9891, \qquad \text{MAE} = 13.3304.$$

Thus C8 strongly supports ranking consistency but not score-level interchangeability.

2 Historical C3 context

Table 1: Latest C3 descriptive statistics.

Year	n	Median	q95	Max
2019	1	5.000	5.000	5.000
2020	1	50.000	50.000	50.000
2021	2	3.500	4.850	5.000
2022	7	12.000	33.500	35.000
2023	13	12.000	29.000	35.000
2024	2673	21.100	38.582	52.080
2025	1902	23.480	41.249	47.220

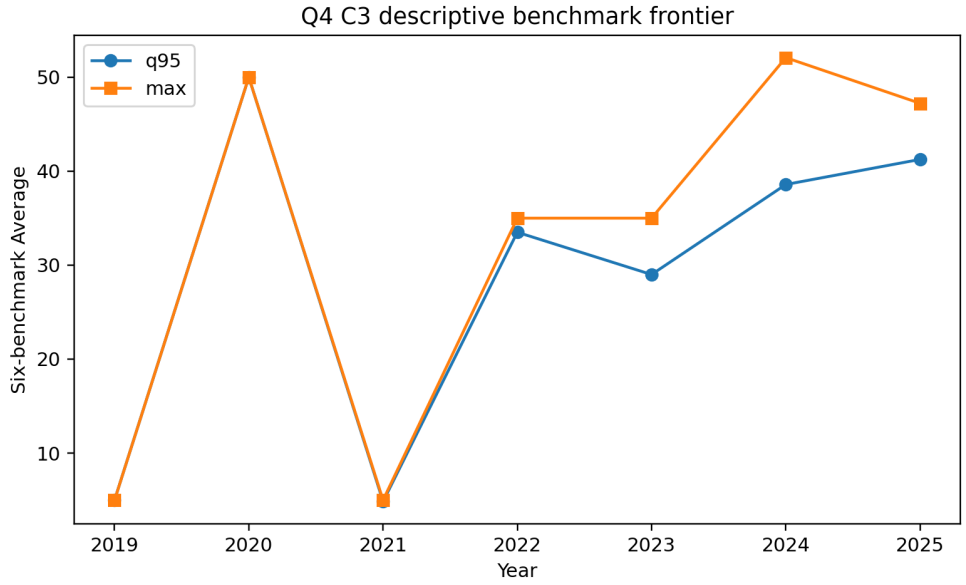


Figure 1: C3 descriptive q95 and maximum benchmark frontier.

Pre-2024 rows are mixed paper/report observations and are not directly comparable with the later leaderboard population.

3 Historical decomposition and specification sensitivity

The primary exact-match sample has 63 deduplicated records and 22 families:

$$S_i = \beta_0 + \beta_C \log_{10} C_i + \beta_t t_i + \epsilon_i.$$

Table 2: Latest decomposition specifications.

Specification	β_C	β_t	R^2	Interpretation
Primary no FE	+4.6796	+4.1279	0.4985	continuity specification
Family FE	+9.3332	-1.9478	0.8498	robustness specification

Primary 95% intervals are $\beta_C \in [1.285, 8.074]$ and $\beta_t \in [1.707, 6.549]$. The family-FE sign flip of β_t means the residual calendar-time term is not robustly identified and cannot be interpreted as a causal technology contribution.

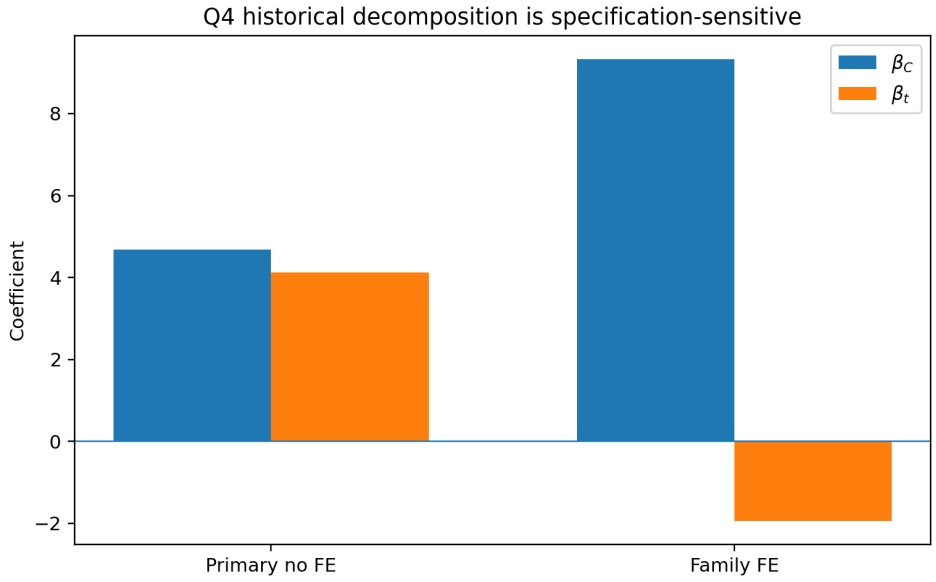


Figure 2: Family fixed effects materially change the decomposition.

Random 5-fold validation gives $R^2 = 0.4345$, MAE 6.1105; family GroupKFold gives $R^2 = 0.3417$, MAE 6.6215. A 2024 out-of-time holdout cannot be formed because the primary matched sample has zero pre-2024 training rows and 56 2024 test rows.

With the dynamically loaded compute-trend slope $g_C = 0.534026$, the primary scale-associated point share is

$$\boxed{37.7\%}, \qquad 95\% \text{ bootstrap} = [4.6\%, 71.7\%].$$

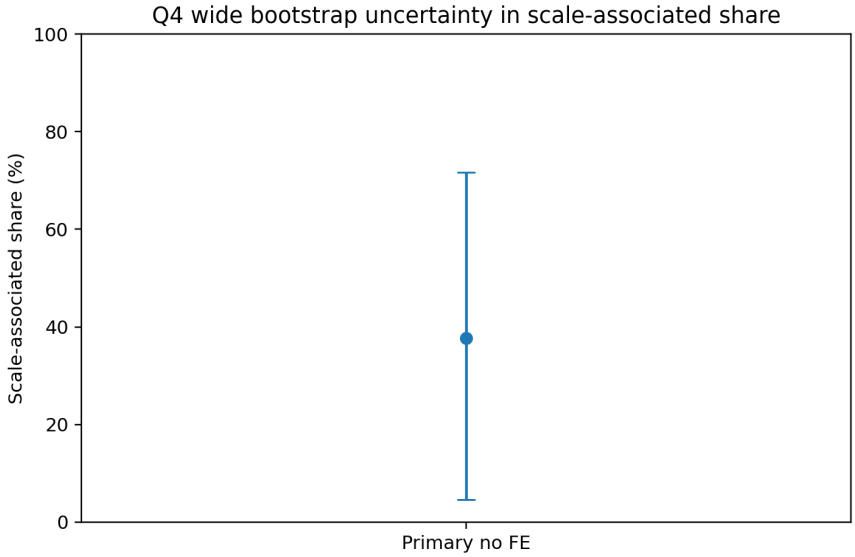


Figure 3: Wide uncertainty in the observational scale-associated share.

4 Compute-frontier trend and documented baseline

The direct forecast anchor is derived from C2 rather than hand-entered: 4,564 valid rows overall, 1,779 rows inside the documented two-year open-license window, latest leaderboard date 2025-03-13, and top model `dfurman/CalmeRys-78B-Orpo-v0.` dated 2024-09-24. The baseline is

$$\boxed{S_0 = 51.2313}.$$

For compute growth, C4 leaves 496 open language models with compute and dates and 31 rolling quarterly anchors:

$$\log_{10} q95(C) = 21.8910 + 0.5340t, \qquad R^2 = 0.9507.$$

HAC(3) gives $[0.4550, 0.6131]$ and the four-quarter block bootstrap gives $[0.4651, 0.5942]$.

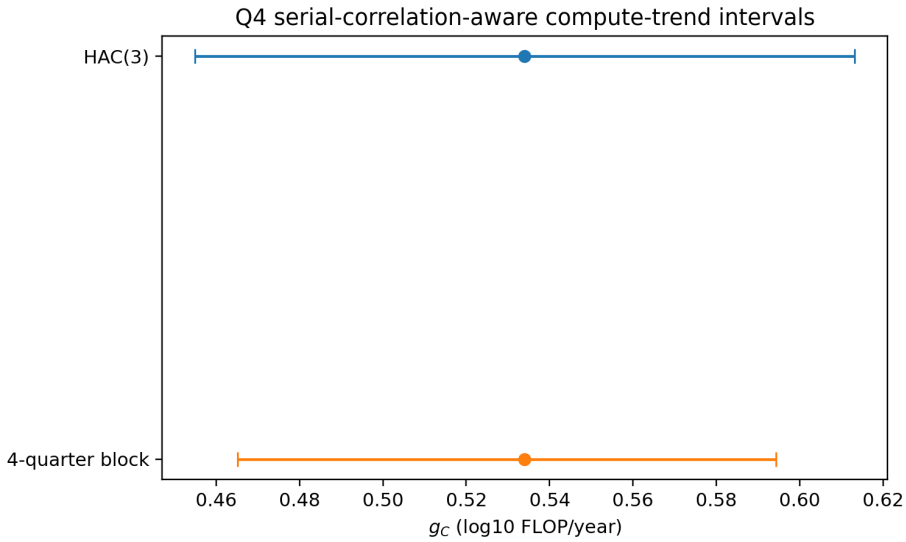


Figure 4: Serial-correlation-aware intervals for the compute-frontier slope.

5 Primary 12/24-month benchmark-space forecast

The headline model is direct benchmark-space extrapolation:

$$\boxed{S(h) = 51.2313 + [\beta_C(sg_C) + \beta_t]h},$$

where s is the retained fraction of recent compute-growth slope and h is years.

Table 3: Primary benchmark-space point scenarios.

Scenario	12 months	24 months
Reference recent growth	57.86	64.49
50% slowdown	56.61	61.99
75% slowdown	55.98	60.74
Compute frontier frozen	55.36	59.49

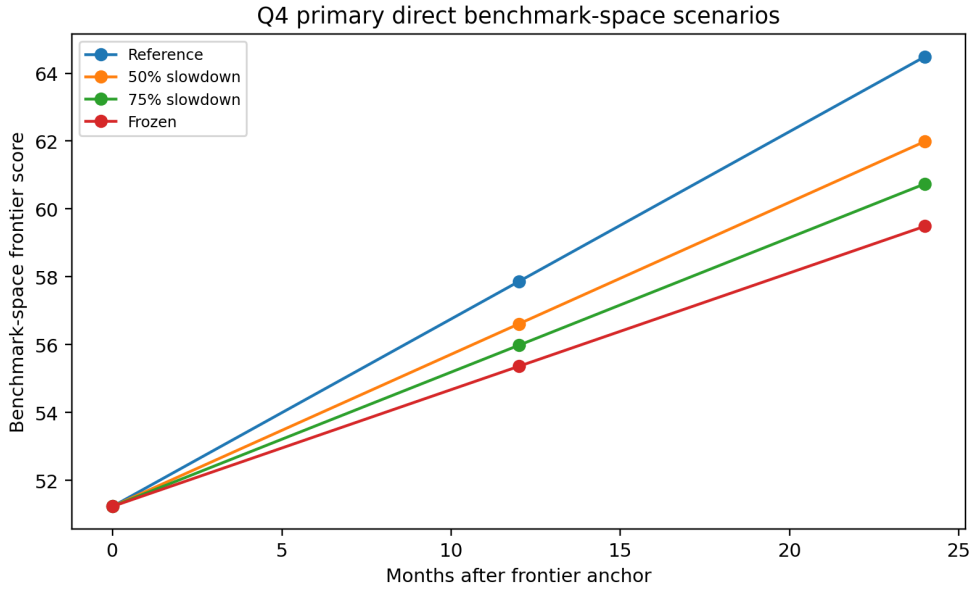


Figure 5: Primary direct benchmark-space scenario projections.

These are conditional point scenarios, not full predictive intervals.

6 Loss-to-Benchmark bridges: direction-only diagnostics

C6 contains 43 rows: seven high-comparability Pythia rows and 36 medium-comparability rows. Two parallel bridge models are retained only as diagnostics.

Table 4: C6 bridge validation.

Model	In-sample R^2	Random 5-fold R^2	GroupKFold R^2	High-weight/high-only R^2
Isotonic (step 04)	≈ 0.25	0.3133	-0.0824	0.3499
Ridge (step 07)	0.2464	0.1043	-0.2429	0.0461

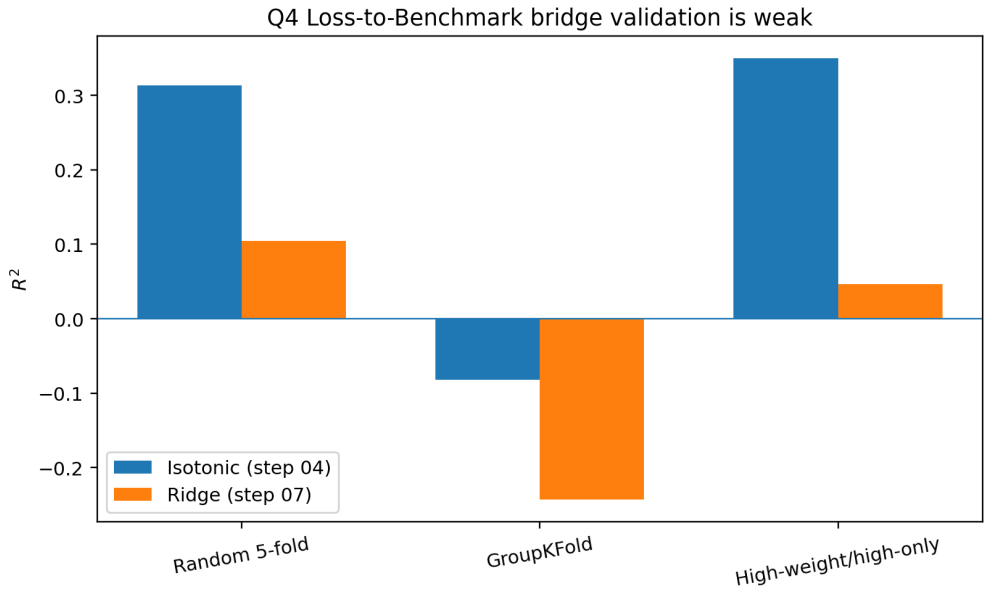


Figure 6: Both bridge forms generalize poorly by model family.

The ridge bridge is $LB = 43.6853 - 14.4118\,L$. Bootstrap predictions from the latest Q3 main solutions are:

Table 5: Secondary Q3-loss to benchmark mapping.

Budget	Q3 loss	Predicted benchmark	95% low	95% high
10^{19}	2.7205	4.4778	0.7911	14.6252
10^{22}	1.9871	15.0476	9.3281	24.0495
10^{24}	1.8026	17.7069	10.7982	27.4030

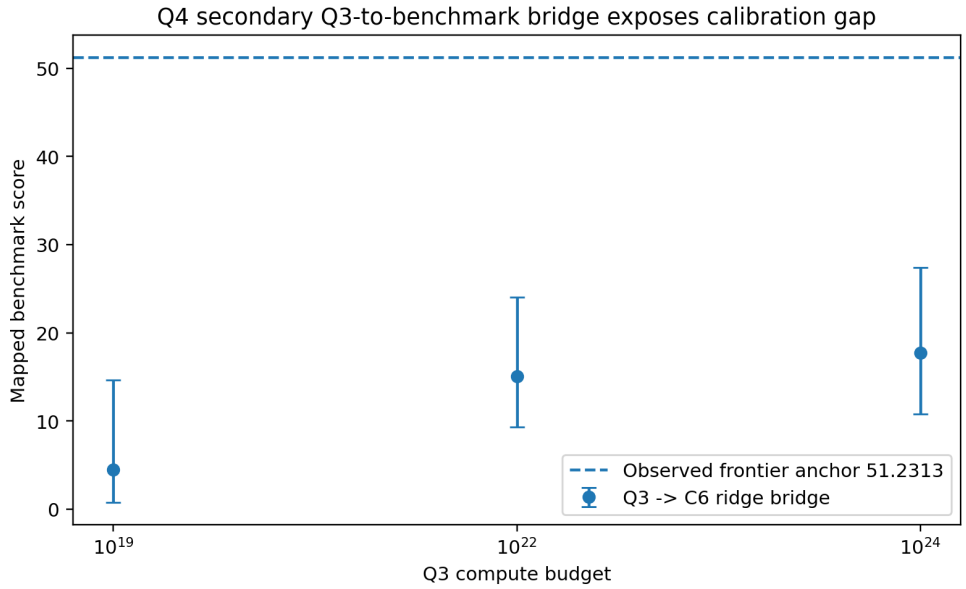


Figure 7: The secondary bridge is far below the observed frontier anchor, exposing its calibration limitation.

At 10^{22} the bridge base is 15.0476 versus direct frontier baseline 51.2313, so it is not used as the headline forecast engine.

6.1 Ridge variant sensitivity

The additional diagnostic `models/q4/bridge_variants_comparison.json` shows:

Table 6: Ridge bridge variants (LOOCV sensitivity).

Variant	n	Slope	LOOCV R^2	MAE	Q3 predictions $10^{19}/10^{22}/10^{24}$
Global all rows, $w = 5$	43	-14.41	+0.073	7.34	4.48 / 15.05 / 17.71
Local within Q3 loss ±0.1	40	-16.01	+0.089	7.52	3.74 / 15.49 / 18.44
High-comparability only	7	-0.14	-0.334	0.38	5.60 / 5.70 / 5.73

The local variant improves LOOCV R^2 by only +0.016 while worsening MAE, so it is not adopted. The seven-point high-only variant collapses the slope and has negative LOOCV R^2; it is also not adopted.

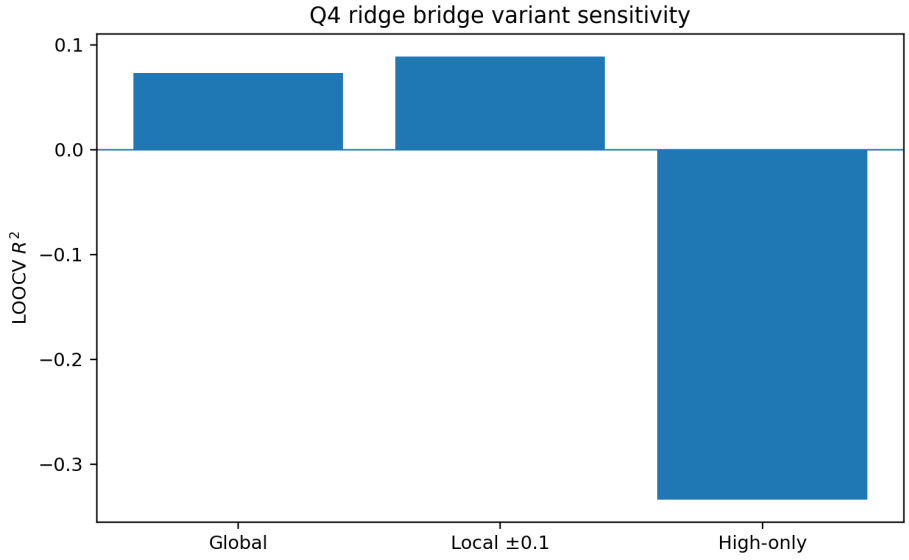


Figure 8: Ridge bridge variant sensitivity.

For completeness, using the global ridge bridge as a secondary cross-check gives a bridge-space base 15.0476

and 12/24-month scores: reference 21.67/28.30, 50% slowdown 20.43/25.80, 75% slowdown 19.80/24.55, and frozen compute 19.18/23.30. These are **not** headline predictions.

7 Known limitations - honest disclosure

Q4 limitations. (1) The Loss-to-Benchmark bridge is weak and family-general calibration is not identified: only 43 C6 rows exist, including seven high-comparability Pythia points. Both isotonic and ridge bridges are direction-only diagnostics. (2) Historical decomposition is specification-sensitive: family fixed effects flip β_t from +4.13 to -1.95, so the scale/non-scale split is observational and not causally robust. (3) The intended 2024 out-of-time holdout has zero pre-2024 training rows and is skipped. (4) The **primary** forecast is directly in benchmark-score space; the bridge-based route is secondary only. (5) The baseline 51.2313 is documented from the two-year C2 open-license window; compute growth itself is estimated from 496 C4 models and 31 rolling anchors. (6) Forecast scores are point scenarios, not full predictive intervals. (7) C8 statistics are computed after repository-key de-duplication.

8 Final Q4 answer

The reproducible primary result is a short-horizon benchmark-space scenario analysis, not a causal forecast. Under the primary no-FE specification and a 50% slowdown in recent compute-frontier growth, the frontier is 56.61 after 12 months and 61.99 after 24 months. The main credibility result is equally important: both the historical attribution and any Loss-to-Benchmark bridge are too specification- and family-sensitive to support precise causal interpretation.